Gadi Medha Spandana

cs24btech11025@iith.ac.in | spandanamedha@gmail.com | +91 6302051856 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.04
Narayana EM HS Dhanalakshnipuram,Nellore	Class XII (board of intermediate education,Andhra pradesh)	2024	97.50%
Narayana Junior College Narasimhakonda,Nellore	Class X (board of secondary education,Andhra pradesh)	2022	96.40%

SCHOLASTIC ACHIEVEMENTS

Secured admission offers to IIT Roorkee and IIT Kharagpur based on JEE Advanced rank; chose to enroll at IIT Hyderabad

PROJECTS

DISHA Platform -- Software Engineering Intern, IIT Hyderabad *(ongoing)*

- Built and shipped backend features across a multi-repo microfrontend platform (Node.js/TypeScript microservices, React MFEs) handling department-based access control for templates and workflows
- Extracted reusable constants and refactored createTemplate/updateTemplate logic across template-service, reducing duplicated department-handling code
- Designed and automated a CI/CD-integrated database seed pipeline, eliminating manual data setup for new environments
- Migrated a custom dropdown component across 22 consumer files to a shared internal package, diagnosing and fixing a Tailwind config issue
- Diagnosed a workflow-inbox visibility bug affecting HOD-level approvals by tracing it to a scope-mismatch in department-based routing

Chat with PDF -- Retrieval-Augmented Generation System

- Built a RAG pipeline using PyMuPDF for text extraction and keyword-based similarity search, integrated with the Gemini API for contextual Q&A

xv6 Operating System -- Kernel Syscalls & Concurrency

- Implemented new kernel syscalls (sys_getppid, sys_getnumchild, sys_getsyscount, sys_getchildsyscount) in a Unix-like teaching OS, working directly with shared kernel data structures.
- Applied correct locking discipline (acquiring locks before reading shared proc-table fields) to avoid race conditions across concurrent processes.

MNIST Digit Classifier -- Neural Network from Scratch (PyTorch)

- Implemented forward/backward propagation and gradient descent without high-level model APIs, achieving 97.6% test accuracy

RELEVANT COURSES

Computer Science: Data Structures & Algorithms, Operating Systems, Database Management Systems

Mathematics: Probability & Statistics

SKILLS

Languages: Python, TypeScript, JavaScript, C

AI/ML: PyTorch, RAG pipelines, prompt engineering (Gemini, Groq APIs), ChromaDB,Pandas, Matplotlib

Web/Backend: React, Node.js, microfrontend architecture (Module Federation), REST APIs, MongoDB, Redis, RabbitMQ

CS Fundamentals: Data structures & algorithms, operating systems, DP, graph algorithms

POSITIONS OF RESPONSIBILITY

Overall Coordinator, Qurve , IIT Hyderabad : Led club operations and produced design/marketing collateral for club events

Core Member, Quiz Club, IIT Hyderabad

EXTRACURRICULAR

Design & Creative Work

- Designed posters and marketing visuals for Qurve events
- Created campaign posters for a friend's General Secretary election campaign, contributing to a successful win
- Participated in NSS-organized creative competitions like clay making, face painting, and t-shirt painting

Community Service & Volunteering

- Volunteered for Office of Career Services (OCS) events at IIT Hyderabad
- Participated in NSS walkathons and campus cleanliness drives